The Ceiling on Skill in Head-to-Head Fantasy Football

A variance budget for a fourteen-team league, 2016–2025

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Abstract

We decompose weekly team scores in a fourteen-team fantasy football league into persistent manager quality and week-to-week noise, and derive from that decomposition the probability that the better manager wins a given matchup. Across 770 manager-weeks the intraclass correlation is $\widehat{\rho} = 0.039$, 95% CI [0.003, 0.065]: roughly *ninety-six percent* of the variation in a weekly score is within-manager noise. The implied skill-to-noise ratio is $\kappa = 0.143$, so a manager one standard deviation above average wins a given week with probability 0.557 and finishes a fourteen-game season above .500 with probability 0.566. Measuring a manager to a reliability of 0.80 would take about ninety-nine weeks, or seven seasons. Substituting each manager's *optimal* lineup for his actual one leaves the decomposition essentially unchanged ($\rho = 0.038$), implying that lineup selection contributes almost nothing to the between-manager variance. Two decisions do carry measurable weight — avoiding early-round draft busts, and waiver claim volume — and we quantify both. We take the modest position that these results describe one league over four seasons and are consistent with, rather than proof of, a general ceiling.

1 Introduction

Ask a group of experienced fantasy football managers how much of the game is skill and you will get two incompatible answers, often from the same person. One is that the game is mostly luck. The other is that some managers make the playoffs every year and this cannot be coincidence. Both claims are usually offered without a number attached.

The two are in fact compatible, and the reconciliation is quantitative. If persistent differences in manager quality are small relative to week-to-week scoring noise, then a real and stable talent advantage will still produce a win rate close to a coin flip in any single week, and a season of fourteen games will be far too short to reveal it. The question is not whether skill exists but what its *magnitude* is on the scale that decides matchups.

This paper estimates that magnitude for one league. Our contributions are:

1. A variance decomposition of weekly team scores into persistent manager quality and within-manager noise, with bootstrap intervals (Section 4.1).

2. A closed-form map from that decomposition to the probability that the better manager wins, and hence to the distribution of season records (Section 3.2).

3. An estimate of the observation horizon required to measure manager quality at conventional reliability (Section 4.3).

4. A decomposition of the three decisions a manager controls — the draft, in-season acquisition, and lineup selection — showing which carry weight and which do not (Section 4.4).

5. An honest validation exercise in which a record-based estimate of the talent spread is compared against the score-based one, and does not fully agree (Section 4.6).

Related work

Becker and Sun [2016] formulate fantasy football draft and lineup management as a mixed-integer program and propose a bilinear model for weekly scoring in which a player's expected production against a given opponent is the product of an innate ability u_i^s for statistic s and the opponent's innate ability to defend it, w_j^{Ds}, fitted by alternating least squares. Two of their results bear directly on the present paper. First, their Table 3 reports mean absolute weekly forecast errors of 6.77 at running back against a mean of 7.46, and 4.61 at tight end against a mean of 4.72; that is, the typical miss is between 87% and 98% of the typical score. Second, in a simulation in which their optimizer is granted *perfect* knowledge of the season's outcomes, it wins a ten-team league only 24.7% of the time against a 10% baseline. Omniscience is worth a factor of 2.5, not a factor of ten.

Lutz [2015] applies support vector regression and neural networks to six seasons of quarterback data and reports a best mean absolute error of 6.22 fantasy points and a mean relative error of 0.42. His machine-learning approaches do not materially outperform the linear baselines he compares against.

The tier lists published at `borischen.co` are widely used in draft preparation and are generated by model-based clustering (Gaussian mixtures via `mclust`) over FantasyPros expert consensus rankings.[1] We note that these tiers cluster expert *opinion*, not realized outcomes, and that the repository publishes no validation of whether tier boundaries carry predictive content.

Our contribution is complementary to all three. Where Becker and Sun [2016] and Lutz [2015] ask how well weekly scores can be predicted, we ask what the irreducible residual implies for the attainable edge, and we work backwards from outcomes rather than forwards from projections.

2 Data

The league is a fourteen-team, half-PPR, head-to-head league with a starting lineup of QB / RB / RB / WR / WR / TE / FLEX (RB, WR, TE) / WRRB_FLEX (RB, WR) / K / DEF and four bench slots. It has run since 2016, on MyFantasyLeague from 2016 to 2019, on Yahoo in 2020–21, and on Sleeper since 2022.

All primary analysis uses the Sleeper era, 2022–2025, obtained from the public Sleeper API: every draft pick, every weekly matchup with the full starter list and per-player point totals, every roster, and every transaction including *failed* waiver claims with their bid amounts. This yields 770 manager-weeks across 14 managers, 798 draft picks, and 1,089 reconstructable waiver auctions.

Analysis is restricted to weeks 1–14, the fantasy regular season. Playoff weeks are excluded because they are not played by all managers and decide nothing for a team already eliminated.

The earlier eras are used only in Section 4.5. The 2016–17 MyFantasyLeague drafts are recoverable; 2018 returns picks with an empty player field, 2019 returns no draft results, and the Yahoo export for 2020–21 contains standings and matchups but no player-level or draft data. The MFL seasons are also structurally different — twenty franchises drafting as two *independent* ten-team drafts from a shared player pool, fifteen rounds, no flex — and MFL stored no scoring rules, so player values there are computed in half-PPR and may not match what that league actually scored.

[1] Source at `https://github.com/borisachen/fftiers`.

3 Methods

3.1 Variance components

Let Y_{mt} be the points scored by the starting lineup of manager m in week t. We posit the one-way random effects model

$$Y_{mt} = \mu + \alpha_m + \varepsilon_{mt}, \qquad \alpha_m \sim (0, \tau^2), \quad \varepsilon_{mt} \sim (0, \sigma^2), \tag{1}$$

with α_m independent of ε_{mt}. Here α_m is the persistent component of manager m's scoring — everything that carries from week to week, including roster quality inherited from the draft — and ε_{mt} is the week-specific deviation. The intraclass correlation

$$\rho = \frac{\tau^2}{\tau^2 + \sigma^2} \tag{2}$$

is the share of variance in a single weekly score attributable to who the manager is.

We deliberately do *not* interpret α_m as "skill" in a causal sense. It absorbs anything persistent, including a lucky draft whose value persists all season. It is an upper bound on skill, not a measurement of it.

3.2 From variance to win probability

A head-to-head matchup is decided by the margin

$$D_{mm't} = Y_{mt} - Y_{m't} = \underbrace{(\alpha_m - \alpha_{m'})}_{\delta} + \underbrace{(\varepsilon_{mt} - \varepsilon_{m't})}_{\text{noise}}. \tag{3}$$

The talent gap δ enters once; the noise enters *twice*, with variance $2\sigma^2$. Assuming approximate normality of the residuals,

$$\Pr(m \text{ beats } m' \mid \delta) = \Phi\left(\frac{\delta}{\sigma\sqrt{2}}\right). \tag{4}$$

It is convenient to express a manager's talent in units of the between-manager standard deviation. Writing $\delta = c\,\tau$ for a manager c standard deviations above average facing an average opponent, and defining the **skill-to-noise ratio**

$$\kappa \;\equiv\; \frac{\tau}{\sigma\sqrt{2}} \;=\; \sqrt{\frac{\rho}{2(1-\rho)}}, \tag{5}$$

equation (4) becomes simply

$$\Pr(\text{win a given week}) = \Phi(c\,\kappa). \tag{6}$$

The second equality in (5) follows by substituting $\tau^2 = \rho(\tau^2 + \sigma^2)$ and $\sigma^2 = (1-\rho)(\tau^2 + \sigma^2)$, and shows that κ is a function of the ICC alone. Everything downstream — win rates, season records, the observation horizon — follows from ρ.

Season records follow immediately. With n independent matchups and per-week win probability $p = \Phi(c\kappa)$, the win total is $W \sim \text{Bin}(n, p)$ and

$$\Pr(\text{winning record}) = \sum_{w > n/2} \binom{n}{w} p^w (1-p)^{n-w}. \tag{7}$$

3.3 Estimation

The panel is unbalanced (managers enter and leave), so we estimate (1) by unbalanced one-way ANOVA [Searle et al., 1992]. With k managers, n_i observations for manager i, and $N = \sum_i n_i$,

$$\widehat{\sigma}^2 = \mathrm{MS}_W, \qquad \widehat{\tau}^2 = \max\left\{\frac{\mathrm{MS}_B - \mathrm{MS}_W}{n_0},\ 0\right\}, \qquad n_0 = \frac{1}{k-1}\left(N - \frac{\sum_i n_i^2}{N}\right). \tag{8}$$

The correction by n_0 matters: the naive between-group variance of the manager means was 26.2, of which $\widehat{\sigma}^2/\bar{n} = 9.5$ is sampling noise in the means themselves. Failing to subtract it would overstate the persistent component by roughly a third.

Interval estimates come from a nonparametric bootstrap resampling *managers* rather than manager-weeks, since observations are clustered within manager; $B = 2000$ replications.

4 Results

4.1 The variance budget

Table 1 gives the decomposition.

	actual lineup		optimal lineup	
	estimate	95% CI	estimate	95% CI
σ^2 (within manager, week to week)	499.5	[450, 550]	520.0	[480, 561]
τ^2 (between managers)	20.3	[1.2, 36.9]	20.6	[0.0, 41.7]
ρ (ICC)	0.0390	[0.0026, 0.0653]	0.0380	[0.0000, 0.0710]
κ (skill-to-noise)	0.1425	[0.0358, 0.1869]	0.1406	[0.0000, 0.1955]
$\Phi(\kappa)$ (P(+1 SD wins a week))	0.557	[0.514, 0.574]	0.556	[0.500, 0.578]

Table 1: Variance budget for weekly team scores, $k = 14$ managers, 770 manager-weeks, 2022–2025. "Optimal lineup" replaces each manager's actual starters with the best legal lineup available from his roster that week.

Two things stand out. First, $\rho = 0.039$: about **96% of the variation in a weekly score is within-manager noise.** In natural units, $\widehat{\sigma} = 22.4$ points against $\widehat{\tau} = 4.5$ points, and the matchup margin carries a noise standard deviation of $\sigma\sqrt{2} = 31.6$ points against a talent gap standard deviation of $\tau\sqrt{2} = 6.4$ points.

Second, and less expected: replacing every manager's actual lineup with his *optimal* lineup leaves the decomposition almost exactly where it was (ρ moves from 0.039 to 0.038). Managers do differ in lineup efficiency — the observed range is 91.1% to 94.1% of optimal — but the differences are too small, relative to σ, to register in the between-manager variance. Perfect start/sit decisions for everyone would not change who wins this league.

4.2 Win probability and season records

Applying (6) and the binomial:

The predicted win rate for a genuinely good manager, 55.7%, is worth comparing against folk estimates. A widely-upvoted claim on `r/fantasyfootball` from a manager reporting his own long-running league's records is that "bad" owners win at 40% and "good" owners at 55–60%. Our figure is derived from the variance decomposition without reference to any observed win rate,

talent edge	Pr(win a week)	$\mathbb{E}$[wins in 14]	Pr(record > .500)
+0.5 SD	0.528	7.40	0.480
+1 SD	0.557	7.79	0.566
+2 SD	0.612	8.57	0.725
+3 SD	0.665	9.32	0.848

Table 2: Implied win probabilities. A manager three standard deviations above average — a figure with no counterpart in the observed data — still loses a third of his weeks.

and lands inside that interval. It is also consistent with Becker and Sun [2016]: an optimizer with perfect foresight won their ten-team simulation 24.7% of the time, against a 10% baseline — a 2.5-fold edge, of the same modest order as ours.

4.3 How long it takes to measure a manager

The reliability of a k-week mean is given by the Spearman–Brown relation $\rho_k = k\rho/(1 + (k-1)\rho)$:

weeks k	14	28	56	104	208
seasons	1.0	2.0	4.0	7.4	14.9
reliability ρ_k	0.362	0.532	0.695	0.809	0.894

Reaching a reliability of 0.80 requires roughly **99 weeks, or seven seasons**. A single season's record has a reliability of 0.36 — barely better than a coin. This is the formal statement of why season-long fantasy standings feel arbitrary: they are.

4.4 Where decisions actually matter

The variance budget bounds the total attainable edge but says nothing about where it lives. We decompose the three decisions a manager controls.

The draft. Each pick is scored as the player's half-PPR points in weeks 1–14, converted to points above replacement at his own position (replacement recomputed each season by greedily filling all fourteen teams' starting slots), minus the mean value returned by picks within ± 12 of that pick number. The between-manager spread concentrates almost entirely in the first four rounds: the three best drafters returned +29.3 points per pick in rounds 1–2 against −28.0 for the three worst, a gap that shrinks monotonically to +4.7 versus −4.6 by rounds 11–14. The mechanism is bust avoidance rather than gem-finding — the top three busted an early pick 8.3% of the time against a league rate of 16.9%. We note that the bust threshold is a free parameter; the *ordering* of managers is stable across thresholds from −100 to 0, but the magnitude of the gap is not.

In-season acquisition. Sleeper retains failed waiver claims with their bid amounts, permitting each contested player-week to be reconstructed as a first-price sealed-bid auction. Across fourteen managers, the correlation between claims attempted per season and waiver production is $r = +0.888$; between price paid per successful claim and production, $r = -0.799$; and between claim *win rate* and production, $r = -0.369$. The negative sign on win rate is the informative one: a high win rate indicates bidding only when certain, and certainty is expensive.

Restricting to claims made in weeks 1–6, so that acquisition date does not confound the comparison, players won for $0 returned 75.7 points and players won for $35 or more returned 78.4 — a 3.6% difference for an unbounded price. Points per dollar fall from 29.9 in the $1–4 band to 1.7 above $35. Since the FAAB budget expires worthless at season end and bench slots recycle weekly, the marginal cost of an additional low bid is close to zero, and the optimal policy is therefore many small claims rather than few large ones.

Lineup selection. As shown in Table 1, this contributes nothing detectable. The entire league sits between 91.1% and 94.1% of the optimal lineup; one standard deviation of lineup efficiency is about 2.8 points per week against a matchup noise standard deviation of 31.6.

4.5 Persistence across eras

If manager quality is a stable trait, draft performance should correlate across time. Comparing each manager's standardised draft score in 2016–17 against 2022–25 for the twelve managers present in both, the correlation is $r = -0.157$, 95% CI $[-0.67, +0.46]$ by Fisher's z. Eight of twelve changed sign; the manager with the best four-year draft record in the current era had the worst in the earlier one.

This estimate is attenuated by measurement error in both halves. With within-era ICC 0.215 for season draft scores, Spearman–Brown gives reliabilities of 0.523 for a four-season mean and 0.354 for a two-season mean, so the disattenuated correlation is $-0.157/\sqrt{0.523 \times 0.354} = -0.365$. Correcting for attenuation moves the estimate *further* from zero in the negative direction; it cannot rescue a claim of persistence. To have supported $r_{\text{true}} = +0.50$ we would have needed to observe $r_{\text{obs}} = +0.215$.

We hold this result loosely. The eras differ in league size, roster structure, and possibly scoring, and four years separate them.

4.6 Validation, and where it fails

The model in Section 3.1 is fitted to weekly *scores*. It makes a testable prediction about season *records*, which are a different measurement. If season win rates are $\text{Bin}(14, p_m)/14$ with $p_m = \Phi(c_m\kappa)$, then their variance decomposes into a binomial part and a talent part. Using the delta method, the talent-induced standard deviation of the win rate is approximately $2\phi(0)\kappa$.

observed SD of season win rate	0.1517
SD implied by a pure coin flip, $n = 14$	0.1336
residual attributable to talent, $\sqrt{0.1517^2 - 0.1336^2}$	0.0718
predicted from the weekly model, $2\phi(0)\kappa$	0.1137

The two do not agree. The weekly-score model predicts a talent spread in season records about 1.6 times larger than the records themselves display. The discrepancy is not fatal — $\widehat{\tau}^2$ has a 95% interval of [1.2, 36.9], and the record-based figure of 0.0718 corresponds to a τ near the lower end of that interval — but it means the point estimate in Table 1 should be read as an *upper* bound. Both measurements agree that the talent effect is small; they disagree by a factor of 1.6 on how small, and we cannot resolve that with four seasons. Candidate explanations include non-normality in the margin (heavier tails reduce the leverage of a mean shift) and correlation in opponent quality induced by the schedule.

5 Limitations

One league. Everything here describes fourteen managers over four seasons. The variance budget may differ in shallower leagues, where replacement level is higher and roster differences larger.

α_m **is not skill.** It is everything persistent, including a draft that broke well. Attributing it to decision quality is an over-reading.

Non-independence. The same managers recur across seasons and face each other; matchups within a week are not independent draws. The bootstrap clusters on managers, which addresses part but not all of this.

Normality. Equation (4) assumes an approximately normal margin. Weekly team scores are right-skewed sums of skewed player scores, and the validation failure in Section 4.6 is consistent with this assumption being imperfect.

Free parameters. The -50 bust threshold and the ± 12 window in the slot-expectation curve were chosen without formal justification. The slot curve is also fitted on the same picks it scores, inducing mild leakage; a leave-one-out or isotonic fit would be preferable.

6 What a manager should conclude

The results support three practical statements and refute a fourth.

First, **the attainable edge is real but small**, on the order of five to six percentage points of weekly win probability for a genuinely good manager. Over fourteen games this is worth about 0.8 wins. It is not nothing — it is roughly the difference between making and missing the playoffs — but it will not feel like control.

Second, **what edge exists is concentrated in the first four rounds of the draft and in waiver volume**. Those are the two decisions with magnitude. The draft evidence is the weaker of the two, resting on sixteen early picks per manager; the waiver evidence rests on 1,089 auctions and is considerably more solid.

Third, **the correct waiver policy is many small claims**, because FAAB is a use-it-or-lose-it budget, bench slots recycle, and price buys almost nothing: a free pickup returned 75.7 points against 78.4 for a $35 claim.

Fourth, and contrary to a great deal of received advice, **weekly start/sit optimisation is not where games are won**. It contributes nothing measurable to between-manager variance. A manager who never watches a football game and simply starts the higher projection while covering bye weeks gives up very little.

The honest summary is that this is a game in which a good manager should expect to be right slightly more than half the time, and in which a single season cannot distinguish him from a lucky one. That is not a reason to stop playing. It is a reason to stop treating the standings as a verdict.

References

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